UNIT 7 · STANDARD 6.EE.6

Write Equations

Lesson 7-1

Name: ______________ Date: ______________ Class: ______________

Key Vocabulary **Level 2 Standard**

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$x + 2 = 7$

balanced with =

$x + 5 = 12$ means 'some number plus 5 equals 12'

Equation

Write the definition:

____

____

x

stands for a number

In $n + 3 = 10$, the letter n stands for the unknown number ($n = 7$)

Variable

Write the definition:

____

____

$x + 2 = 7$

balanced with =

$7 + 3 = 10$ — both sides equal 10

Equal sign

Write the definition:

____

____

$3x + 5$

no equal sign

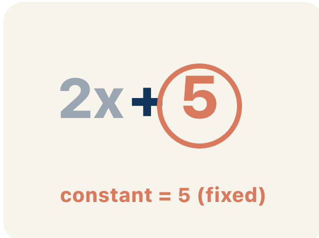
$2x + 5$ is an expression; it becomes an equation when you write $2x + 5 = 15$

Expression

Write the definition:

____

____



In x + 7 = 12, the numbers 7 and 12 are constants; only x can change.

Constant

Write the definition:

____

____

In n + 5 = 12, n is the unknown; you solve to find n = 7.

Unknown

Write the definition:

____

____

Guided Notes

Level 2 Standard

🎯 **WHAT WE'RE LEARNING TODAY**

I can write an equation to represent a real-world situation.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Equation | **Variable** | **Equal sign** | **Expression** | **Constant** | **Unknown**

1. A math sentence that says two amounts are equal, with an = sign, is an ____________ .

2. A letter that stands for an unknown number is a ____________ .

3. The symbol = that shows two amounts are the same is the ____________ .

4. A group of numbers and variables with no equal sign is an ____________ .

5. A fixed number that does not change is a ____________ .

6. The value you are trying to find in an equation is the ____________ .

🧮 Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 I do — watch

Follow each step as your teacher solves it.

Problem: Which equation represents: 'A number plus 12 equals 30'?

A. $n + 12 = 30$

B. $n - 12 = 30$

C. $12n = 30$

D. $n / 12 = 30$

Step 1 'Plus' means addition, so the equation is $n + 12 = 30$.

✅ **Answer:** A. $n + 12 = 30$

🤝 We do — together

Solve this one with your class using the same steps.

Problem: Which equation represents: 'Seven less than a number is 18'?

A. $n - 7 = 18$

B. $7 - n = 18$

C. $n + 7 = 18$

D. $n / 7 = 18$

Step 1 ______________________________

Step 2 ______________________________

Answer: ______________________________

✏️ You do — your turn

Now try one on your own. Show every step.

Problem: Which equation represents: 'Three times a number equals 21'?

A. $3n = 21$

B. $n + 3 = 21$

C. $n - 3 = 21$

D. $n / 3 = 21$

Show your work:

____

____

____

Try It

Solve on your own. Check the answer key when you are done.

1. The lab tech says: '5 times the number of evidence bags b equals 35.' Which equation represents this clue?

A. 5b = 35

B. b + 5 = 35

C. b − 5 = 35

D. b / 5 = 35

Show your work:

____

____

____

2. A torn note reads: 'A suspect's case minutes m, after 9 fewer, leaves 24.' Which equation fits?

A. m − 9 = 24

B. m + 9 = 24

C. 9m = 24

D. m / 9 = 24

Show your work:

____

____

____

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Write a real-world situation that can be modeled by the equation n / 5 = 9. Explain what n represents and verify your equation makes sense.

Sentence starter: Situation: ____. In this problem, n represents ____. Check: n = ____ because ____ ÷ 5 = 9.

Show your work:

Reflect — Exit Ticket

Which equation represents: 'A number divided by 6 equals 9'?

A. n / 6 = 9

B. 6n = 9

C. n − 6 = 9

D. n + 6 = 9

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Equation
2. **Notes 2:** Variable
3. **Notes 3:** Equal sign
4. **Notes 4:** Expression
5. **Notes 5:** Constant
6. **Notes 6:** Unknown
7. **We Try (notes):** A. $n - 7 = 18$ — *'Seven less than a number' means start with the number and subtract 7: $n - 7 = 18$.*
8. **You Try (notes):** A. $3n = 21$ — *'Three times' means multiply by 3, so the equation is $3n = 21$.*
9. **Try It 1:** A. $5b = 35$ — *'5 times the number' means 5 times b, so $5b = 35$. Check: $5 \times 7 = 35$.*
10. **Try It 2:** A. $m - 9 = 24$ — *'9 fewer' means subtract 9, so $m - 9 = 24$. Check: $33 - 9 = 24$.*
11. **Exit Ticket:** A. $n / 6 = 9$ — *'Divided by 6' means division: $n / 6 = 9$. Check: $54 / 6 = 9$ ✓*